

NSCLC-Radiogenomics — participant flow (CLAIM item 14)

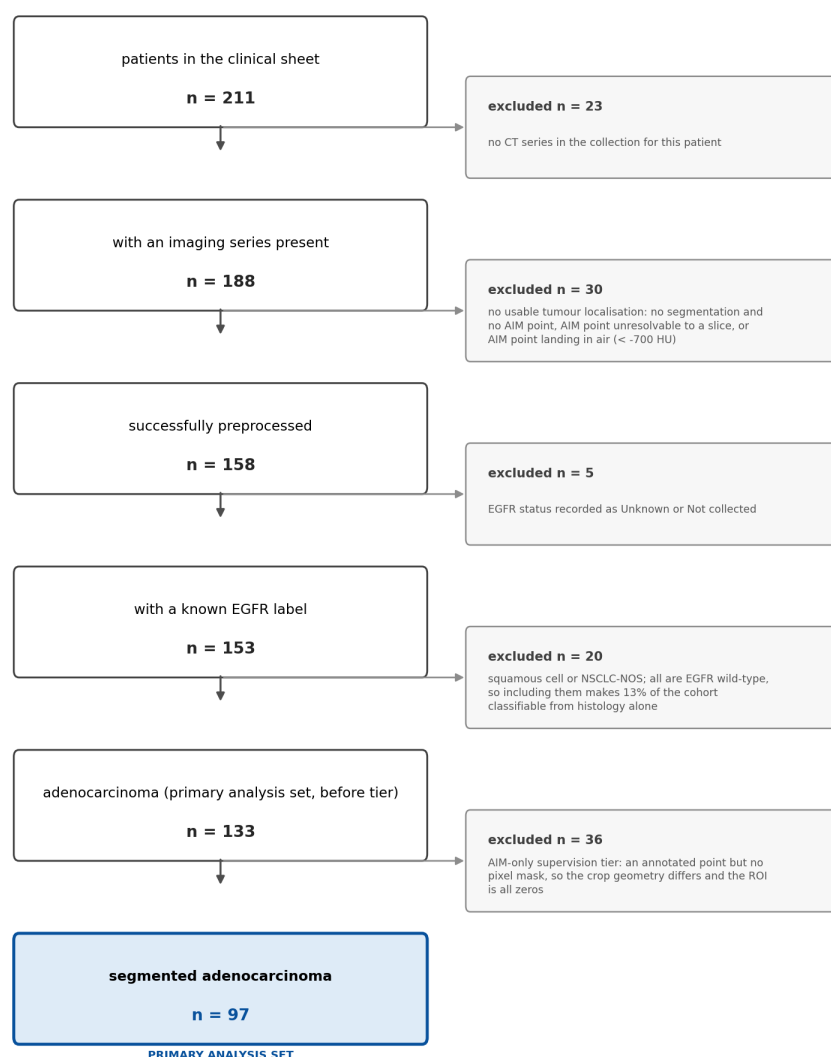


Figure 1: Flowchart shows participant flow through the study. Counts and exclusion reasons are generated from the analysis pipeline rather than transcribed, and reconcile exactly $(211 - 114 = 97)$. The pre-specified primary analysis set is patients with segmented adenocarcinoma. EGFR = epidermal growth factor receptor, NSCLC = non-small cell lung cancer.

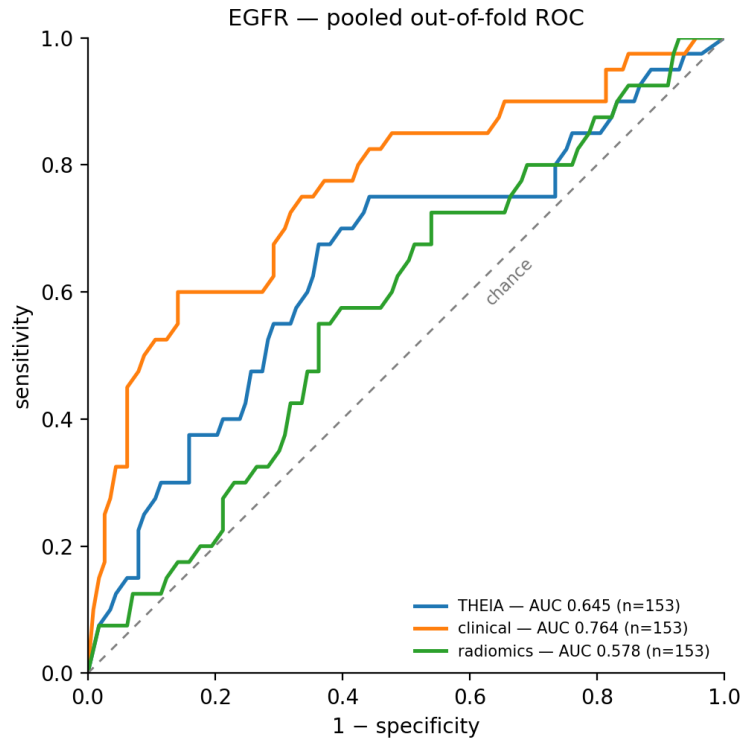


Figure 2: Graph shows pooled out-of-fold receiver operating characteristic curves by arm, computed on identical patients and folds. The chance line is drawn and labeled, and each arm's number of patients is printed because the arms do not all cover the same patients. AUC = area under the receiver operating characteristic curve.

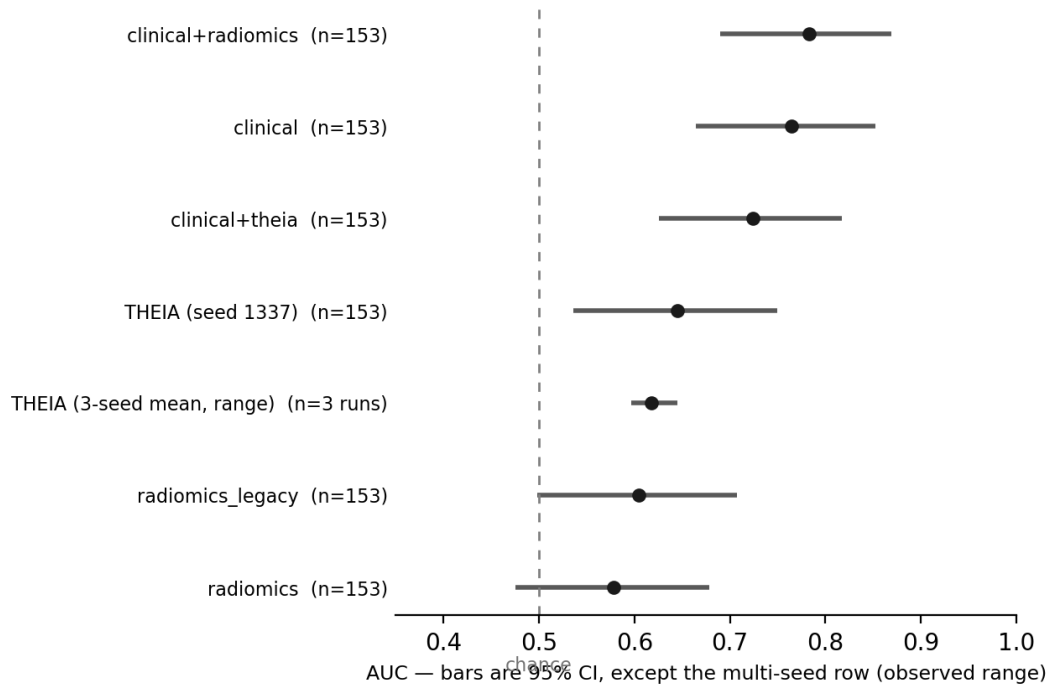


Figure 3: Forest plot shows every arm with its 95% confidence interval. Overlapping intervals can still differ significantly when the comparison is paired, which is why the paired differences are reported alongside. AUC = area under the receiver operating characteristic curve, CI = confidence interval.

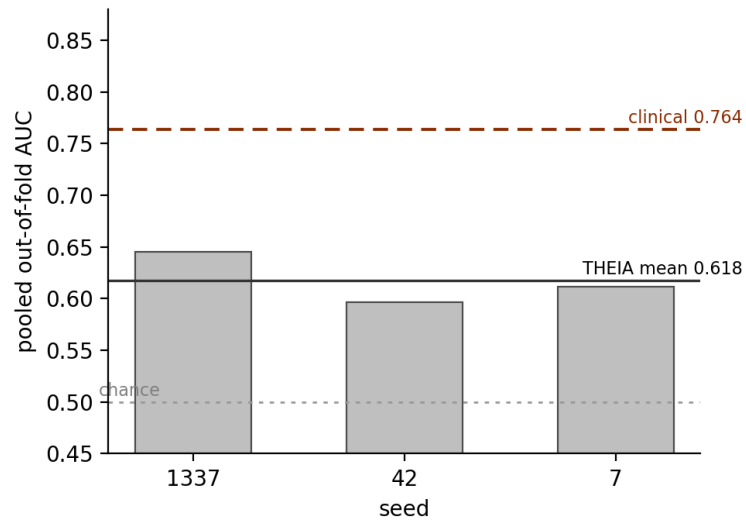


Figure 4: Graph shows per-seed area under the receiver operating characteristic curve against the clinical baseline. The rerun spread is shown next to the effect being claimed; the gap to a five-variable clinical model exceeds it. AUC = area under the receiver operating characteristic curve.

External validation: 420 held-out NSCLC-Radiomics patients (Maastricht, NL) — mass lift +0.244, beats shuffle 11/15

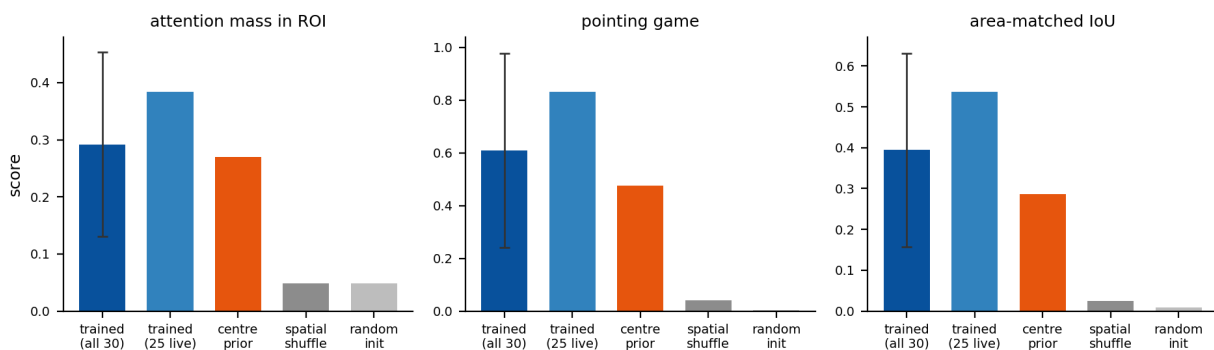


Figure 5: Graphs show the primary endpoint — localization on 420 held-out NSCLC-Radiomics patients — with all three controls in every panel. The center prior is strong on attention mass by construction and weak on pointing accuracy and area-matched intersection over union, which are invariant to its width, so the trained model’s margin is read from the right-hand panels rather than the left. IoU = intersection over union, NSCLC = non-small cell lung cancer.

attention (hot) with tumour outlined (green) — first 6 masked patients of held-out fold 0, not selected on score

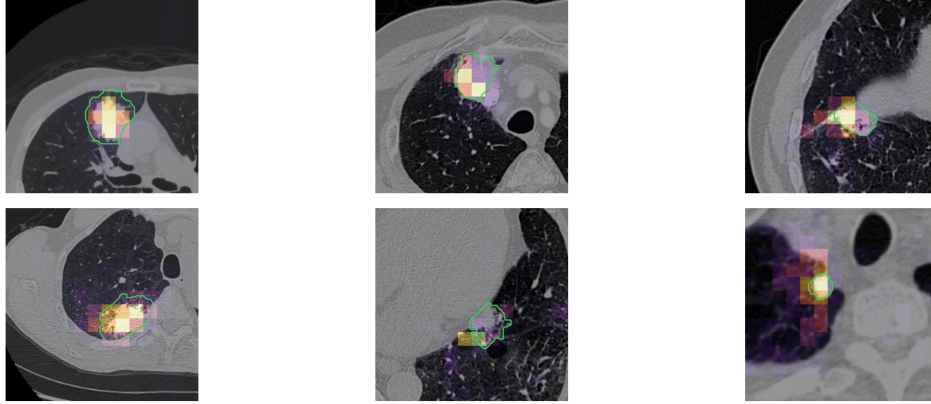


Figure 6: CT images with superimposed attention maps (hot colors) and the tumor contour drawn from the reference segmentation (green) in six patients: (A) 68-year-old man with adenocarcinoma, (B) 76-year-old man with NSCLC not otherwise specified, (C) 64-year-old man with NSCLC not otherwise specified, (D) 63-year-old man with adenocarcinoma, (E) 84-year-old man with adenocarcinoma, and (F) 59-year-old man with squamous cell carcinoma. These are the first six patients with a segmentation, in index order, from held-out fold 0 of training run B in Table 3; they were not selected on score, and the histologic mix is whatever that rule produced. Images are from the NSCLC-Radiogenomics collection, used under its CC BY 3.0 license (9). NSCLC = non-small cell lung cancer.